Purview Message Encryption (OME) Implementation — Microsoft Purview

Implementation Solution Guide

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Executive Summary — Purview Message Encryption (OME) in Microsoft Purview

The problem

Email is still the main way sensitive information leaves an organization — contracts, payroll, health and financial details, account numbers — and by default it leaves **unprotected**. Once a message reaches an external recipient it can be forwarded, printed, copied, and stored anywhere, with no control and no record. That is a compliance gap and a data-risk exposure at the same time: the information is out, and there is no way to restrict or account for what happens to it.

The solution

This solution implements **Purview Message Encryption** — the current **Office 365 Message Encryption (OME)**, built on **Azure Rights Management** — so that outbound email can be **encrypted, branded, and usage-restricted**, to any recipient, on any device:

- **Encrypt** — the message is protected in transit and at rest; the recipient can read and reply.
- **Do Not Forward** — the message is encrypted **and** the recipient **cannot forward, print, or copy** it, for the most sensitive sends.
- **Your branding** — the encryption experience carries **your** logo, colour, and text, so recipients trust it rather than mistaking it for phishing.
- **Automatic, rule-driven** — mail flow rules apply the right protection when a message matches a condition (a keyword, a sensitivity label, an external recipient), so protection doesn't depend on a user remembering.

How it works

We follow a repeatable model — **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand** — that enables the **encryption foundation first** (Azure RMS, proven healthy), sets the **branding**, then builds the mail flow rules in **Audit mode scoped to a pilot group**, where they **match and log but do not yet encrypt**. Only after sanctioned test sends prove the experience end to end — encrypted, branded, readable by internal and external recipients, and (for Do Not Forward) genuinely non-forwardable — do the rules move to **Enforce**, and scope widens deliberately.

The solution ships as **self-guided documentation plus five PowerShell tools** (Exchange Online / Security & Compliance PowerShell / Graph) that assess readiness and RMS health, configure branding and the rules (Audit-first), validate the wiring, and produce a leadership rollup — all **read-only or `-WhatIf`/Audit-mode by default**, with **no credentials in code**.

Why it matters

- **Sensitive email is protected** — encrypted and, where needed, non-forwardable, instead of leaving in the clear.

• **Recipients trust it** — a branded experience that reaches anyone, inside or outside the organization.

• **Automatic and defensible** — rule-driven encryption with an evidence trail (message trace), piloted before it enforces, so nothing breaks silently.

• **A fraction of consulting cost** — the same disciplined method a specialist would use, delivered as a repeatable, tested runbook.

Important boundaries

OME is **licensing-dependent**: **Basic OME** (via Azure Information Protection, included in common E3/E5 plans) provides encryption, the two templates, and a **single default** branding; **custom branding templates, message revocation, and expiration** require **Advanced Message Encryption** (Microsoft 365 E5 / E5 Compliance) — capability availability and exact portal paths/parameters should be **verified against your licensing and tenant**. Deciding **what is sensitive** — which mail must be encrypted and what triggers Do Not Forward — is a **compliance, privacy, and legal determination** for your organization; this solution provides the technical method, the pilot discipline, and the evidence, not the determination.

Solution Overview — Purview Message Encryption (OME) in Microsoft Purview

What this solution delivers

Email is still the way most sensitive information leaves an organization — a contract, a payroll file, a patient summary, a set of account numbers — and by default it leaves **in the clear**. Once a message crosses to an external recipient it can be forwarded, printed, copied, and retained anywhere, with no control and no evidence of who did what. That is a compliance gap and a data-risk exposure at the same time. **Purview Message Encryption** — the current, "new" **Office 365 Message Encryption (OME)**, built on **Azure Rights Management (Azure RMS)** — closes it: outbound email can be **encrypted**, branded with your organization's look, and delivered with a **usage restriction** such as **Do Not Forward (DNF)** or **Encrypt-only**, to recipients inside or outside the tenant, on any device.

This solution walks a competent-but-novice administrator through a **complete, repeatable OME deployment**:

1. **Turn on the encryption foundation** — confirm/enable **Azure RMS** for the tenant (`Set-IRMConfiguration -AzureRMSLicensingEnabled $true`) and prove it healthy with `Test-IRMConfiguration`. No RMS, no OME.

1. **Brand the encryption experience** — configure the **OME branding** (the encryption portal / wrapper email your recipients see) with your logo, colours, and text via `Set-OMEConfiguration` (the default configuration), and — where licensing allows — a **custom branding template** via `New-OMEConfiguration` for specific flows.

1. **Make encryption happen automatically with rules** — build **mail flow (transport) rules** that

apply the `Encrypt` or `Do Not Forward` protection template when a message matches a condition (a keyword, a sensitivity label, an external recipient, a classification), **piloted in the rule's `Audit` mode and scoped to a pilot group first**, so nothing encrypts in production until you have proven the match.

1. **Give users the manual controls too** — the **Encrypt** and **Do Not Forward** options in Outlook

/ Outlook on the web, so a person can protect a one-off message deliberately.

1. **Validate, then expand** — prove the end-to-end path (message → rule match → encrypted, branded,

restricted delivery → recipient reads it) with sanctioned test sends, then widen scope and flip the rules from **Audit** to **Enforce** deliberately.

Together these turn **plaintext outbound email** into **encrypted, branded, usage-restricted** email — configured, piloted, validated, and defensible.

OME, the two templates, and where the pieces live — read this first

• **OME is built on Azure RMS.** The "new" OME encrypts the message and controls what the recipient

can do with it (view, reply, forward). It is delivered through **Exchange Online** — so the configuration cmdlets live in **Exchange Online PowerShell** (`Connect-ExchangeOnline`), **not** Security & Compliance PowerShell. That is deliberate and correct for this capability; the scripts here connect to Exchange Online (and, for label-driven encryption, Security & Compliance PowerShell).

• **The two built-in protection templates.**

• `Encrypt` — the message is encrypted; the recipient can read, **reply, and forward** (the

protection travels, but usage is not restricted). Good for "keep it confidential in transit and at rest, but let them work with it."

• `Do Not Forward` **(DNF)** — the message is encrypted **and** the recipient **cannot forward,

print, or copy** it. Good for the most sensitive one-to-one or one-to-few sends.

• **Encrypt-only** is a lighter, forward-permitting option surfaced to users in Outlook.

• **Branding is a separate object.** The **OME configuration** (`Get/Set/New-OMEConfiguration`)

controls the *look* of the encryption experience — the wrapper email text, the portal logo, background colour, and disclaimer. The **default** configuration ("OME Configuration") brands every OME send; **custom** branding templates (multiple, per-flow) require **Advanced Message Encryption** (see licensing below).

• **Rules apply the templates.** A **mail flow (transport) rule** with the action *Apply Office 365

Message Encryption and rights protection* (parameter `ApplyRightsProtectionTemplate` = `Encrypt` or `Do Not Forward`, optionally with a custom branding template) is what makes encryption **automatic** based on a condition. Rules have a `Mode` — `Audit`, `AuditAndNotify`, or `Enforce` — which is our pilot lever: build in `Audit` first (the rule matches and logs but **does not encrypt**), then flip to `Enforce`.

> ***Determinations are the business's, not the tool's.*** **Which* messages must be encrypted, *what* classification triggers DNF, and *how long* protected mail may be readable are* ***compliance/privacy/legal determinations*** *for your organization (see* `global-requirements-purview.md` *§5). This solution delivers the* ***technical method, the pilot discipline, and the evidence*** *— not the policy decision about what is sensitive.*

Basic OME vs Advanced Message Encryption — licensing shapes what you get

OME capability is **licensing-dependent** — verify against your tenant:

- **Basic OME** (Azure RMS via **Azure Information Protection**, included in Microsoft 365 E3/E5, Office 365 E3/E5, Microsoft 365 Business Premium, EMS E3/E5, AIP Plan 1): encryption, the **Encrypt** and **Do Not Forward** templates, and a **single default** branding configuration.
- **Advanced Message Encryption** (Microsoft 365 E5, Office 365 E5, Microsoft 365 E5 Compliance / Information Protection & Governance): **multiple custom branding templates**, plus, for mail sent to external recipients via the OME portal, **message revocation** (`Set-OMEMessageRevocation`) and **expiration**.

> ***Verify availability against your licensing/tenant.*** *Custom branding templates, revocation, and expiration are **Advanced** features; if you only have Basic OME, plan on the **default** branding and the two templates, and treat revocation/expiration as "verify your licensing."*

The standard deployment model

Every step follows the same repeatable model, with an explicit **pass/fail gate** between phases:

Assess → Design → Configure → Pilot (Audit mode / pilot group) → Validate → Expand (Enforce).

- **Assess** — confirm licensing (Basic vs Advanced OME), Exchange admin roles, that **Azure RMS is enabled and healthy** (`Test-IRMConfiguration`), the in-scope trigger conditions, and compliance/privacy sign-off on *what* gets encrypted.
- **Design** — decide the branding, the rule conditions, the template per condition (`Encrypt` vs `Do Not Forward`), the pilot scope, and name every object.
- **Configure** — enable RMS, set the branding, then build the mail flow rules in `Audit` **mode**, scoped to a **pilot group**.
- **Pilot** — send sanctioned test mail; confirm the rules **match** (Audit logs / message trace) **without encrypting production mail**.
- **Validate** — run the pass/fail gates in `05-testing.md` (RMS health, branding renders, template applied, recipient experience, DNF actually blocks forwarding).
- **Expand** — flip the rules from **Audit** to **Enforce**, then widen scope deliberately, re-running the gates.

What's in the box

- This overview, a **Requirements** guide, an ordered **Implementation** guide, **Testing** (pass/fail gates), **Troubleshooting**, **Best Practices**, and a **Deliverables Summary** + **Executive Summary**.
- **Five PowerShell tools** (Exchange Online PowerShell / Security & Compliance PowerShell / Graph):

a **discovery/readiness** assessment, a **configuration/remediation** helper for branding and the mail flow rules (`-WhatIf`/Audit-first), a **validation** checker, an **encryption reporting** rollup, and a **scope/health** helper — **read-only or `-WhatIf` by default; no credentials in code**.

> ***Sources:*** *Microsoft Learn — *Message encryption* / *Office 365 Message Encryption*, *Set up new Office 365 Message Encryption capabilities*, *Add branding to encrypted messages* (`Get/Set/New-OMEConfiguration`), *Define mail flow rules to encrypt email messages* (`ApplyRightsProtectionTemplate`, rule `Mode`), *Manage Azure Information Protection / IRM for Exchange Online* (`Set-IRMConfiguration`, `Test-IRMConfiguration`), and *Advanced Message Encryption* (custom templates, revocation, expiration). OME is **licensing-dependent** (Basic OME via AIP; Advanced Message Encryption for custom branding / revocation / expiration) and the Purview/Exchange portals **evolve — verify capability availability and exact paths/parameters against your licensing/tenant** rather than assuming.*

Requirements — Purview Message Encryption (OME) in Microsoft Purview

Confirm every prerequisite below **before** you configure anything. Category-level prerequisites (tenant, compliance licensing model, least-privilege roles, S&C PowerShell / Graph toolchain, in-scope data sources, compliance/privacy/legal stakeholders, pilot scope + change control) are in `global-requirements-purview.md` — reference them; they are **not** repeated here. The items below are the ones **specific to OME**: the **Azure RMS foundation**, the **Exchange Online** administration surface, the **branding** object, the **mail flow rules**, and the licensing that separates **Basic** from **Advanced** message encryption.

Each requirement is written as **Description · Why · Verify · Implement if missing.**

> ***Note the toolchain nuance.*** *OME is delivered through **Exchange Online**, so its cmdlets (`Set-IRMConfiguration`, `*-OMEConfiguration`, `*-TransportRule`) live in **Exchange Online PowerShell** (`Connect-ExchangeOnline`), not Security & Compliance PowerShell. Label-driven encryption uses S&C PowerShell (`Connect-IPPSSession`). Both are in the ExchangeOnlineManagement module (`global-requirements-purview.md` §3).*

R1. Licensing for OME — Basic (AIP/Azure RMS) vs Advanced Message Encryption

Description: For **Basic OME** (encryption, the `Encrypt` and `Do Not Forward` templates, a single **default** branding), the **Azure Information Protection / Azure RMS** entitlement included in Microsoft 365 E3/E5, Office 365 E3/E5, Microsoft 365 Business Premium, EMS E3/E5, or AIP Plan 1, assigned to the **senders** whose mail will be encrypted. For **Advanced Message Encryption** (multiple **custom** branding templates, message **revocation**, and **expiration** for external recipients), Microsoft 365 E5 / Office 365 E5 / Microsoft 365 E5 Compliance (Information Protection & Governance). **Why:** OME rides on Azure RMS; the template and default branding are Basic, but custom branding / revocation / expiration are **Advanced-only**. Senders must be licensed for the protection to apply. **Verify:** Microsoft 365 admin center → **Billing** → **Licenses** — confirm the AIP/RMS entitlement and, for Advanced, the E5/E5 Compliance SKU. `OME-Discovery.ps1` reports RMS enablement and flags whether custom-branding/revocation features are in play. Cross-check the *Microsoft Purview / Azure Information Protection service

description* — **verify against your licensing/tenant. Implement if missing:** Acquire/assign the AIP/RMS entitlement for Basic OME; add E5/E5 Compliance where custom branding, revocation, or expiration are required. Do **not** design around Advanced features you are not licensed for.

R2. Azure Rights Management (Azure RMS) enabled and healthy for the tenant

Description: Azure RMS **activated** for the tenant and **enabled for Exchange Online IRM** (`Set-IRMConfiguration -AzureRMSLicensingEnabled $true`), passing `Test-IRMConfiguration`. **Why:** OME cannot encrypt anything if the RMS foundation is off or unhealthy — every template, rule, and branding depends on it. **Verify:** Exchange Online PowerShell — `Get-IRMConfiguration` shows `AzureRMSLicensingEnabled : True`; `Test-IRMConfiguration -Sender <addr> -Recipient <addr>` returns **PASS** on all checks. `OME-Discovery.ps1` / `OME-ScopeHealth.ps1` report this. **Implement if missing:** In Exchange Online PowerShell run `Set-IRMConfiguration -AzureRMSLicensingEnabled $true`, then re-run `Test-IRMConfiguration`. (Azure RMS activation itself is in the Microsoft Purview / Azure portal — **verify the current path in your tenant.**)

R3. Exchange Online administration roles (least privilege)

Description: An admin with the rights to configure **IRM**, **OME configuration**, and **mail flow (transport) rules** — the Exchange Online **Organization Management** role, or a least-privilege custom role group that includes the **Transport Rules** and **IRM/message-encryption** management roles. Global Administrator only where genuinely required (e.g. initial Azure RMS activation). **Why:** `Set-IRMConfiguration`, `*-OMEConfiguration`, and `*-TransportRule` are Exchange Online admin actions; without the matching role they fail. **Verify:** Exchange admin center → **Roles**, or `Get-ManagementRoleAssignment` for the account. `OME-Discovery.ps1` reports whether the connected account can read the relevant configuration. **Implement if missing:** Assign the least-privilege Exchange role group under change control; reserve Global Administrator for the one-time RMS activation only.

R4. The encryption branding decision (default configuration, and any custom templates)

Description: The branding your recipients should see on the OME experience — the wrapper email text, portal **logo**, **background/accent colour**, **disclaimer**, and "read the message" text — set on the **default** OME configuration, plus a decision on whether any **custom** branding templates (Advanced only) are needed for specific flows. **Why:** The encryption portal/wrapper is a recipient-facing, trust-critical surface; unbranded or default-Microsoft branding undermines confidence and looks like phishing. Custom templates need **Advanced Message Encryption** (R1). **Verify:** Exchange Online PowerShell — `Get-OMEConfiguration` shows the default configuration and any custom ones. `OME-Discovery.ps1` lists existing OME configurations. **Implement if missing:** Gather the approved logo (per Microsoft's size/format guidance — **verify current limits**), colours, and text; set them on the default configuration (`OME-Set-...` / `Set-OMEConfiguration`). Only create custom templates if licensed and a flow genuinely needs distinct branding.

R5. The trigger conditions and template mapping (what gets encrypted, and how)

Description: The **conditions** that should cause encryption (e.g. a subject/keyword like `[encrypt]`, a **sensitivity label**, an **external** recipient, a classification / sensitive info type) and the **template** each maps to (`Encrypt` vs `Do Not Forward`), agreed with compliance/privacy. **Why:** The mail flow rule keys off these conditions; getting them wrong either **over-encrypts** (breaking legitimate flows / recipient usability) or **under-encrypts** (leaving sensitive mail in the clear). The *what-is-sensitive* determination is the business's (§5), not a technical default. **Verify:** Confirm the agreed condition→template mapping in writing with compliance/privacy before authoring rules. `OME-Discovery.ps1` inventories any existing OME rules. **Implement if missing:** Work with compliance/privacy to define the conditions and template mapping; start narrow (a keyword + external recipient) and add classification/label conditions as they mature.

R6. Sensitivity labels / classification (optional but recommended trigger source)

Description: Where you want encryption driven by **classification** rather than a keyword, the **sensitivity labels** (or sensitive info types) that the rule — or the label's own protection — will key on, published and in use on the in-scope mail. **Why:** Label-driven encryption is more robust than a keyword: a **Confidential** or **Highly Confidential** label can either **apply encryption itself** (label-based protection, which can invoke OME branding for email) or serve as a **mail flow rule condition** that applies a template. **Verify:** Purview → **Information Protection** → **Labels / Label policies** shows the labels published; S&C PowerShell `Get-Label` reads them. `OME-Discovery.ps1` (with S&C session) reports label presence. **Implement if missing:** Publish the needed labels first (this can be a separate labeling project). If you have no labels yet, start with keyword/recipient conditions and add label conditions once labeling is in use.

R7. Recipient experience and supported flows

Description: Awareness of how **different recipients** read protected mail — Microsoft 365 recipients read it inline; other recipients (Outlook.com, Gmail, Yahoo, or arbitrary providers) use a **one-time passcode** or federated sign-in via the **OME portal** — and confirmation that these flows are acceptable for your correspondents. **Why:** OME is designed to reach **any** recipient, but the external experience (portal + OTP) differs from inline reading; setting expectations avoids "the encrypted email doesn't work" support tickets. **Verify:** Confirm the intended external recipient domains and test the OTP/portal flow to one during the pilot (Testing Gate 5). **Verify current supported-recipient behaviour in Microsoft docs. Implement if missing:** Document the recipient experience for your help desk and end-users (`HELP_CENTER`-style guidance) before broad rollout.

R8. A pilot scope and Audit-mode change control

Description: A **named pilot sender group** (`SG-OME-Pilot`) and agreement to build the mail flow rules in `Audit` **mode** (the rule matches and logs but **does not encrypt**) scoped to that group first, under a **change-control** process, before any tenant-wide `Enforce`. **Why:** A mis-scoped encryption rule can encrypt mail that must stay readable by systems/recipients that can't handle OME, or over-restrict with DNF; Audit mode + a pilot group prove the **match** without changing delivery, so you can tune before enforcing. (`global-requirements-purview.md` §6.) **Verify:** Confirm `SG-OME-Pilot` exists, that the rules will be created with `-Mode Audit`, and a change window/approval path. `OME-ScopeHealth.ps1` previews the pilot group and rule scope. **Implement if missing:** Create the pilot group, author rules in `Audit` mode, and agree change control before flipping any rule to `Enforce`.

Global Requirements — Microsoft Purview Solutions

These requirements apply to **every** solution in the **Purview** category (implementing and managing Microsoft Purview, sensitivity & retention labels, DLP, data lifecycle & records, insider risk, communication compliance, audit, data security posture management, and data governance). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every Purview solution).*** *The **Microsoft Purview portal** (purview.microsoft.com), the **Microsoft 365 admin center**, **Security & Compliance PowerShell** (`Connect-IPPSSession`), and **Microsoft Graph** are authored at exact, current click-by-click / cmdlet detail from Microsoft's public documentation. Microsoft Purview features are heavily **licensing-/plan-dependent** (many require **Microsoft 365 E5** or the **E5 Compliance** add-on; some are in preview) and the portal is **actively evolving** — capability availability and portal paths are **cited or flagged "verify your licensing/tenant"**, never asserted or invented. **Regulatory, records-retention, and legal determinations** (retention schedules, records file plans, legal holds, insider-risk case actions, regulatory applicability) are **deferred to the organization's compliance, records, privacy, and legal functions** — the solutions provide the technical configuration method and evidence, not the determination. Automation is **PowerShell** (Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets.*

1. Microsoft 365 tenant with the required Purview/compliance licensing

• **Description:** A Microsoft 365 tenant with the **compliance licensing** the capability requires

(E3 for baseline; **E5 / E5 Compliance** for advanced DLP, auto-labeling, insider risk, communication compliance, Audit Premium, DSPM, etc.).

• **Why required:** Most Purview capabilities are plan-gated; the feature won't be available without the

right plan.

• **How to verify:** In the M365 admin center confirm the subscription/add-ons; check the feature is

visible in the Purview portal.

• **How to implement if missing:** Assign/purchase the required licensing before starting.

2. Purview roles (least-privilege) and portal access

• **Description:** Access with the least-privilege **Purview / compliance role** the solution names

(e.g. Compliance Administrator, Compliance Data Administrator, Information Protection, Records Management, Insider Risk, eDiscovery, Audit roles), with Global Administrator only where required.

• **Why required:** Purview uses fine-grained role groups; configuration requires the matching role.

• **How to verify:** Purview portal → **Roles & scopes / Permissions** — confirm the role membership.

• **How to implement if missing:** Assign the least-privilege role group under change control.

3. Security & Compliance PowerShell + Graph toolchain

- **Description:** A workstation with **Security & Compliance PowerShell** (`Connect-IPPSSession` via the

ExchangeOnlineManagement module) and **Microsoft Graph PowerShell**, modern-auth / MFA-capable admin.

- **Why required:** Configuration, automation, and the provided scripts run through these.
- **How to verify:** `Connect-IPPSSession` and `Connect-MgGraph` succeed with the admin account.
- **How to implement if missing:** Install the modules and confirm modern-auth sign-in before running steps.

4. Data sources & scope in scope

- **Description:** The **workloads and data sources** in scope (Exchange, SharePoint, OneDrive, Teams,

endpoints/devices, and — for data governance — the broader data estate) and a **data-classification / sensitivity** intent to drive labels, DLP, and retention.

- **Why required:** Purview policies key off the in-scope workloads and the classification scheme.
- **How to verify:** Confirm the in-scope workloads and the classification/sensitivity approach with

stakeholders.

- **How to implement if missing:** Define the scope and classification intent before authoring policies.

5. Compliance, records, privacy & legal stakeholders

- **Description:** Engagement with the **compliance, records-management, privacy, security, and legal**

stakeholders whose policies the configuration must satisfy (retention schedules, records file plans, regulatory obligations, insider-risk and communication-compliance policy, legal hold).

- **Why required:** Purview configuration must align to enterprise policy and legal/records requirements,

not technical defaults; several capabilities have HR/legal sensitivity.

- **How to verify:** Confirm the applicable policies, the records schedule, and named approvers.
- **How to implement if missing:** Identify the owners and capture the requirements before configuring.

6. A pilot scope and change control

- **Description:** A **pilot scope** (users/sites/policies in simulation or a limited ring) to validate

on before tenant-wide enforcement, and a **change-control** process — most Purview policies support a **simulation / test / disabled** mode that MUST be used before enforcement.

- **Why required:** Every solution follows an **Assess → Design → Configure → Pilot (simulate) → Validate

→ Expand (enforce)** model; DLP, labels, retention, and insider-risk changes are high-impact.

- **How to verify:** Confirm a named pilot scope, that simulation/test mode is available, and a change

window/approval path.

- **How to implement if missing:** Define the pilot and agree change control before enforcing anything.

*The deployment model across every Purview solution is the same: **Assess** → **Design** → **Configure** → **Pilot (simulate/test)** → **Validate** → **Expand (enforce)**, with explicit pass/fail gates and a documented rollback. Automation is delivered as **PowerShell** (Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets. Regulatory/records/legal determinations are deferred to the relevant governance functions.*

Implementation — Purview Message Encryption (OME) in Microsoft Purview

This is the ordered, do-this-then-that guide. It follows the standard model (**Assess** → **Design** → **Configure** → **Pilot (Audit mode / pilot group)** → **Validate** → **Expand (Enforce)**) and — because a mis-scoped encryption rule changes how mail is delivered — it enables the **RMS foundation first**, sets **branding**, then builds the mail flow rules in `Audit` **mode scoped to a pilot group** before anything encrypts in production. Every object is named verbatim so the scripts, testing, and troubleshooting line up. Replace the `contoso.com` / sample names with your own.

> ***Toolchain note.*** *OME is administered in **Exchange Online** — these steps use **Exchange Online PowerShell** (`Connect-ExchangeOnline`): `Set-IRMConfiguration`, `*-OMEConfiguration`, `*-TransportRule`, `Test-IRMConfiguration`. Label-driven encryption additionally uses **S&C PowerShell** (`Connect-IPPSSession`). Some settings (Azure RMS activation, and the branding preview) are surfaced in the Microsoft Purview / Exchange admin portals — where a path or parameter may differ by tenant/licence, **verify against your licensing/tenant.***
>
> ***Hard gate before you enforce.*** *Do **not** flip any mail flow rule from `Audit` to `Enforce` until compliance/privacy have confirmed **what** should be encrypted and to which template (R5), and the Testing gates pass. *What is sensitive* is a business determination (`global-requirements-purview.md` §5), not a technical default.*

Phase 0 — Assess (confirm you can build safely)

1. **Confirm licensing (R1).** Verify the AIP/Azure RMS entitlement for the pilot senders (Basic OME); confirm E5/E5 Compliance if you need **custom branding / revocation / expiration** (Advanced).
1. **Confirm the RMS foundation (R2).** Connect Exchange Online PowerShell and run `OME-Discovery.ps1` (read-only). Confirm `AzureRMSLicensingEnabled : True` and that `Test-IRMConfiguration` passes.

- **Gate 0a:** RMS is enabled and `Test-IRMConfiguration` passes → continue. Otherwise run `Set-IRMConfiguration -AzureRMSLicensingEnabled $true` (and activate Azure RMS in the portal), then re-test.

1. **Confirm roles (R3).** The connected account holds Exchange **Organization Management** (or the least-privilege transport-rules + IRM role group).

1. **Confirm the toolchain** (`global-requirements-purview.md` §3): `Connect-ExchangeOnline` succeeds

(and `Connect-IPPSSession` if you will use label-driven encryption).

Phase 1 — Design (agree it with compliance/privacy, then name everything)

Decide and **record with compliance/privacy**:

• **Branding (R4):** the default OME configuration's logo, background/accent colour, portal text,

disclaimer, and email text; and whether any **custom** template (`OME-Branding-Corp`, Advanced only) is needed.

• **Trigger conditions → template (R5):**

• `OME-Encrypt-Outbound-Pilot` — condition: sender ∈ `SG-OME-Pilot` **AND** (subject

contains `[encrypt]` **OR** recipient is **external**) → template `Encrypt`.

• `OME-DoNotForward-Confidential` — condition: message carries the `Highly Confidential`

sensitivity label (or the agreed classification/header) → template `Do Not Forward`.

• **Pilot scope:** a named group, `SG-OME-Pilot`, holding a small set of sanctioned pilot senders.

• **Manual controls:** confirm the Outlook **Encrypt** / **Do Not Forward** options are enabled for

users who need to protect one-off messages deliberately.

Phase 2 — Configure the RMS foundation and branding

In **Exchange Online PowerShell**:

1. **Enable RMS (if not already, R2).** `Set-IRMConfiguration -AzureRMSLicensingEnabled $true`, then

`Test-IRMConfiguration -Sender pilot@contoso.com -Recipient external@example.com` — confirm all checks **PASS**.

1. **Set the default branding.** Use `OME-Configure.ps1 -Action Branding -WhatIf` to preview, then

`-Commit` to apply, or run `Set-OMEConfiguration` directly against the default configuration (identity **"OME Configuration"**) — logo, `BackgroundColor`, `EmailText`, `PortalText`, `DisclaimerText`, `ReadButtonText` (**verify current property names in your tenant**).

1. **(Advanced only) Create a custom branding template.** If licensed, `New-OMEConfiguration -Identity

"OME-Branding-Corp" then Set-OMEConfiguration -Identity "OME-Branding-Corp" ...` for a per-flow look.

• **Pass/fail:** `Get-OMEConfiguration` returns the branding you set; a test send renders it (Phase 4 /

Testing Gate 4). **Rollback:** re-run `Set-OMEConfiguration` to restore prior values (record the originals first with `OME-Discovery.ps1`).

Phase 3 — Build the mail flow rules in AUDIT mode, scoped to the pilot

In **Exchange Online PowerShell**, create the rules with `OME-Configure.ps1 -Action MailFlowRules -WhatIf` (prints the exact plan), then `-Commit` (creates them in `Audit` **mode**). The equivalent direct commands:

1. `OME-Encrypt-Outbound-Pilot` (template `Encrypt`, Audit first):

```
New-TransportRule -Name "OME-Encrypt-Outbound-Pilot" `
  -FromMemberOf "SG-OME-Pilot@contoso.com" `
  -SentToScope NotInOrganization `
  -SubjectContainsWords "[encrypt]" `
  -ApplyRightsProtectionTemplate "Encrypt" `
  -Mode Audit
```

(Combine/loosen the `-SubjectContainsWords` / `-SentToScope` conditions to match your Phase-1 design; to apply a custom branding template add `-ApplyRightsProtectionCustomizationTemplate "OME-Branding-Corp"` — **verify the exact parameter name in your tenant**.)

1. `OME-DoNotForward-Confidential` (template `Do Not Forward`, Audit first):

```
New-TransportRule -Name "OME-DoNotForward-Confidential" `
  -FromMemberOf "SG-OME-Pilot@contoso.com" `
  -HeaderMatchesMessageHeader "msip_labels" `
  -HeaderMatchesPatterns "<Highly Confidential label GUID>" `
  -ApplyRightsProtectionTemplate "Do Not Forward" `
  -Mode Audit
```

(The `msip_labels` header carries the applied sensitivity-label GUID — **verify the header name and your label GUID**; alternatively key on a keyword/classification per Phase 1.)

• **Pilot rule:** every rule is created with `-Mode Audit` and scoped to `SG-OME-Pilot`. In `Audit` mode the rule **matches and logs but does not encrypt** — so nothing changes for real mail yet.

• **Pass/fail:** `Get-TransportRule` shows both rules **enabled,** `Mode = Audit`, scoped to the pilot group. `OME-Validate.ps1` confirms mode/scope. **Rollback:** `Disable-TransportRule` (or `Remove-TransportRule`); Audit mode changes nothing for users, so pilot rollback is trivial.

Phase 4 — Pilot and validate (run the gates)

Work the gates in `05-testing.md`. In summary, from a **sanctioned pilot sender** in `SG-OME-Pilot`:

• Send a test message that matches `OME-Encrypt-Outbound-Pilot` (subject `[encrypt]`, external recipient). In `Audit` **mode**, confirm via **message trace** / rule Audit logs that the rule **matched** — without the message actually being encrypted yet.

• Temporarily flip **only the one pilot rule** to `Enforce` for a controlled test window (or use a dedicated test rule) and confirm a real test send is **encrypted, branded, and readable** by an internal and an external test recipient (OTP/portal for external — R7).

• Confirm `Do Not Forward` actually **prevents forwarding/printing/copying** for the recipient.

• Confirm reporting (`OME-Report.ps1`) reconciles with **message trace**.

• **Do not enforce broadly** until every gate passes **and** compliance/privacy confirm the

condition→template mapping (R5).

Phase 5 — Expand (enforce, then widen — only after the gate passes)

1. **Flip rules to Enforce.** Change each rule's mode: `Set-TransportRule -Identity

"OME-Encrypt-Outbound-Pilot" -Mode Enforce` (and the DNF rule). Watch the first cohort closely via message trace and help-desk signal.

1. **Widen scope deliberately** — add members to `SG-OME-Pilot` (or broaden the rule's `-FromMemberOf`

/ `-SentToScope`) **group-by-group**, not tenant-wide in one move, using `OME-ScopeHealth.ps1` (read-only preview) to stage the expansion. **Re-run the gates** after each expansion.

1. **(Advanced) Enable revocation/expiration if licensed.** For external OME-portal mail, confirm

`Get-OMEMessageStatus` and `Set-OMEMessageRevocation` are available, and set expiration on the OME configuration (`ExternalMailExpiryInDays` — **verify property/licensing**).

Rollback summary

Change	Rollback
Branding wrong / broken logo	Re-run `Set-OMEConfiguration` to restore the prior values (captured by `OME-Discovery.ps1` before you changed them).
Encryption rule too broad / breaking a flow	`Set-TransportRule -Mode Audit` (or `Disable-TransportRule`) — encryption stops immediately; tune conditions and re-pilot.
DNF too restrictive for a flow	Change that rule's template to `Encrypt`, or narrow its condition; DNF only where forwarding must be blocked.
Rule scoped too far	Remove members from `SG-OME-Pilot` (`OME-ScopeHealth.ps1` preview first), or tighten `-FromMemberOf`; the rule follows the scope.
RMS/IRM change caused mail issues	Re-run `Test-IRMConfiguration` to isolate; `Set-IRMConfiguration` to correct; RMS enablement is tenant-wide, so change it under change control.

> ***Never** treat OME as a substitute for the business decision about **what is sensitive**. The conditions and templates here **enforce** a policy; deciding *which mail* must be encrypted, and for how long it may be readable, is a **compliance/privacy/legal determination** (§5). This solution provides the method and the evidence, not the determination.*

Testing & Validation — Purview Message Encryption (OME) in Microsoft Purview

Run these gates in order. Each has an explicit **pass/fail** signal. **Do not enforce broadly** (Phase 5 — flip rules from `Audit` to `Enforce` and widen scope) until every gate passes **and** compliance/privacy have confirmed the condition→template mapping. Where a gate involves a real send, use a **sanctioned pilot sender** in `SG-OME-Pilot` and **test recipients** (one internal Microsoft 365 mailbox, one external mailbox on a different provider) — do not test on production correspondence.

Use the scripts where noted; they are read-only or `-WhatIf`/Audit-mode and change nothing you don't approve.

Gate 1 — Licensing is in place (Basic vs Advanced)

Why: OME rides on Azure RMS/AIP; custom branding, revocation, and expiration are **Advanced-only**. **How:** Confirm the AIP/Azure RMS entitlement for the pilot senders; confirm E5/E5 Compliance if you rely on custom branding / revocation / expiration. Run `OME-Discovery.ps1`. **Pass:** Senders are licensed for Basic OME; any Advanced feature you plan to use is licensed. **Fail:** Senders unlicensed, or a design depends on an Advanced feature you don't have → acquire/assign, or redesign to Basic (default branding, two templates) before proceeding.

Gate 2 — Azure RMS is enabled and healthy

Why: No RMS, no encryption — every template, rule, and branding depends on it. **How:** Exchange Online PowerShell — `Get-IRMConfiguration` (`AzureRMSLicensingEnabled : True`) and `Test-IRMConfiguration -Sender <pilot> -Recipient <external>`. `OME-ScopeHealth.ps1` runs these. **Pass:** `AzureRMSLicensingEnabled` is **True** and `Test-IRMConfiguration` returns **PASS** on all checks (RMS online, templates retrieved, encrypt/decrypt round-trip). **Fail:** RMS disabled or any `Test-IRMConfiguration` check fails → enable RMS (`Set-IRMConfiguration -AzureRMSLicensingEnabled $true`) / activate Azure RMS, then re-test.

Gate 3 — The protection templates are available

Why: The rules apply `Encrypt` / `Do Not Forward`; they must exist and be published. **How:** Exchange Online PowerShell — `Get-RMSTemplate` lists the available templates. `OME-Discovery.ps1` reports them. **Pass:** The `Encrypt` and `Do Not Forward` templates are returned and enabled. **Fail:** Templates missing → confirm RMS is active (Gate 2) and allow propagation; **verify template availability in your tenant.**

Gate 4 — Branding is configured and renders correctly

Why: The encryption portal/wrapper is recipient-facing and trust-critical; wrong or missing branding looks like phishing. **How:** `Get-OMEConfiguration` shows the branding you set (logo, colour, text). Then send a sanctioned encrypted test to an internal and an external recipient and **view** the wrapper/portal. **Pass:** `Get-OMEConfiguration` returns your intended values; the recipient sees your logo, colour, and text (not default Microsoft branding, unless that is the deliberate choice). **Fail:** Branding absent, broken (logo too large / wrong format), or wrong → re-run `Set-OMEConfiguration` (Implementation Phase 2) and re-test; **verify current logo size/format limits.**

Gate 5 — The rules are healthy, in AUDIT mode, scoped to the pilot

Why: A prematurely-enforcing or over-scoped rule encrypts (or DNFs) mail that must stay readable. **How:** `Get-TransportRule`; `OME-Configure.ps1 -Action MailFlowRules -WhatIf` to re-print the plan; `OME-Validate.ps1` to confirm mode/scope. **Pass:** `OME-Encrypt-Outbound-Pilot` and `OME-DoNotForward-Confidential` are **enabled**, `Mode = Audit`, scoped to `SG-OME-Pilot`, each with the correct condition and template. **Fail:** A rule is already `Enforce`, over-scoped, disabled, or has the wrong condition/template → correct before testing detections.

Gate 6 — A matching message is detected in AUDIT (without being encrypted)

Why: Proves the condition matches the intended mail **before** anything encrypts. **How:** From a pilot sender, send a message that matches `OME-Encrypt-Outbound-Pilot` (subject `[encrypt]`, external recipient). Review **message trace** / the rule's Audit results. **Pass:** Message trace / Audit shows the rule **matched** the test message; a **non-matching** control message (no `[encrypt]`, internal only) did **not** match. The matching message is **not** encrypted (Audit mode). **Fail:** No match on the test, or a match on the control → fix the rule conditions/scope before enforcing.

Gate 7 — A real encrypted send is encrypted, branded, and readable

Why: Proves the end-to-end path in a controlled window before broad enforcement. **How:** In a controlled test window, flip **only the pilot Encrypt rule** to `Enforce` (or use a dedicated test rule), send a sanctioned message, and open it as **(a)** an internal Microsoft 365 recipient and **(b)** an external recipient (one-time passcode / OME portal — R7). Then return the rule to `Audit` if broad enforcement isn't yet approved. **Pass:** Both recipients can read the message; the external recipient completes the OTP/portal flow; the branding renders. Message trace shows the message was encrypted by the rule. **Fail:** Either recipient cannot read it, the OTP/portal flow fails, or the rule didn't encrypt → check RMS health (Gate 2), recipient support (R7), and the rule action.

Gate 8 — Do Not Forward actually restricts the recipient

Why: DNF's value is that the recipient **cannot** forward/print/copy — this must be proven, not assumed. **How:** Send a sanctioned message via the `Do Not Forward` path to an internal test recipient and attempt to **forward**, **print**, and **copy** it. **Pass:** Forward/print/copy are **blocked/greyed** for the DNF message; the same actions are allowed on an ordinary (`Encrypt`-only) message. **Fail:** The recipient can forward/print a DNF message → confirm the rule applied `Do Not Forward` (not `Encrypt`), and that the recipient client honours RMS restrictions.

Gate 9 — Reporting reconciles with message trace

Why: Leadership needs a defensible summary of what was encrypted, without over-collecting. **How:** Run `OME-Report.ps1` (read-only) and compare its rollup to **message trace** (and, for Advanced external mail, `Get-OMEMessageStatus`). **Pass:** The rollup (encrypted-message counts by rule/template/time) reconciles with message trace; no content or recipient data is exposed beyond what the report is authorized to show. **Fail:** Rollup diverges materially from message trace → message trace is authoritative; reconcile before reporting.

Gate 10 — Compliance/privacy confirm the mapping before broad enforcement

Why: *What* must be encrypted and *what* triggers DNF are business determinations, not technical defaults. **How:** Walk the condition→template mapping with compliance/privacy: which conditions encrypt, which apply DNF, and the recipient experience. **Pass:** Compliance/privacy confirm the mapping and the recipient experience in writing. **Fail:** Ambiguity about what should be encrypted, or the mapping isn't signed off → resolve **before** flipping rules to `Enforce`.

Exit criteria (all must be true before Expand/enforce)

- Licensing in place — Basic (and Advanced where relied on) (Gate 1)
- Azure RMS enabled and healthy; `Test-IRMConfiguration` passes (Gate 2)
- `Encrypt` and `Do Not Forward` templates available (Gate 3)
- Branding configured and renders correctly (Gate 4)
- Rules healthy, `Audit` mode, pilot-scoped (Gate 5)
- Matching message detected in Audit without being encrypted (Gate 6)
- Controlled real send: encrypted, branded, readable internally + externally (Gate 7)
- Do Not Forward restricts forward/print/copy (Gate 8)
- Reporting reconciles with message trace (Gate 9)
- Compliance/privacy confirmed the condition→template mapping (Gate 10)

6. Troubleshooting — Purview Message Encryption (OME) in Microsoft Purview

The failure modes you'll hit configuring and operating OME. Each: **symptom** → **likely cause** → **fix.** Most "encryption isn't working" reports are actually RMS not enabled, a rule still in `Audit` mode, a condition that doesn't match, or a recipient-experience issue — this guide separates those from real faults. OME is administered in **Exchange Online PowerShell** (`Connect-ExchangeOnline`); label-driven encryption also uses **S&C PowerShell** (`Connect-IPPSSession`).

`Test-IRMConfiguration` fails, or `AzureRMSLicensingEnabled` is False

- **Cause A — RMS not enabled.** Azure RMS/IRM is the foundation; without it nothing encrypts. **Fix:** `Set-IRMConfiguration -AzureRMSLicensingEnabled $true`, activate Azure RMS in the portal, then re-run `OME-ScopeHealth.ps1 -Sender <pilot> -Recipient <external>` (Gate 2).
- **Cause B — licensing.** The sender lacks an AIP/Azure RMS entitlement. **Fix:** confirm licensing (Requirement R1); **verify your licensing/tenant** — plan names evolve.
- **Cause C — propagation.** RMS was just enabled. **Fix:** allow propagation, then re-test.

`Get-RMSTemplate` returns no `Encrypt` / `Do Not Forward` template

• **Cause:** RMS isn't active yet, or templates haven't propagated. **Fix:** confirm Gate 2 passes, allow time, re-run `OME-Discovery.ps1`. **Verify template availability in your tenant** — do not invent template names.

A message that should encrypt isn't being encrypted

• **Cause A — rule in Audit mode (expected during pilot).** In `Audit` the rule **matches and logs but does not encrypt **— this is by design until you enforce.** Fix:** confirm with `OME-Validate.ps1`; encryption begins only when the rule is deliberately flipped to `Enforce` (Phase 5, after the gates).

• **Cause B — condition mismatch.** The subject keyword (`[encrypt]`), recipient scope (`NotInOrganization`), or label header/GUID doesn't match the test message. **Fix:** re-check the rule conditions against the test (`OME-Configure.ps1 -Action MailFlowRules -WhatIf` reprints the plan); confirm the sender is in `SG-OME-Pilot`.

• **Cause C — sender out of scope.** The sender isn't a member of `SG-OME-Pilot`. **Fix:** confirm membership with `OME-ScopeHealth.ps1`.

The label-driven Do-Not-Forward rule doesn't fire

• **Cause:** the `msip_labels` header pattern doesn't match your actual label GUID, or the message never carried the label. **Fix:** capture the real header from a labeled test message and confirm the **Highly Confidential** label GUID (**verify the header name and your label GUID in your tenant**); alternatively key on the agreed keyword/classification per the Phase-1 design. *Which* mail warrants DNF is a compliance/privacy determination — configure to their mapping.

External recipient can't open the encrypted message

• **Cause A — recipient experience.** External recipients read via **one-time passcode / OME portal** (Requirement R7); a mail/social account mismatch or a blocked portal breaks the flow. **Fix:** have them use the OTP path; confirm the OME portal is reachable from their side.

• **Cause B — branding renders as untrusted.** Missing/broken branding looks like phishing. **Fix:** re-run `Set-OMEConfiguration` (Phase 2), verify with `OME-Validate.ps1` and a test send (Gate 4); **verify current logo size/format limits.**

Branding doesn't appear, or the logo is broken

• **Cause A — wrong identity.** You set branding on a custom template but the rule uses the default configuration (or vice-versa). **Fix:** confirm which configuration the rule applies; align them.

• **Cause B — logo format/size.** The image exceeds current limits or is an unsupported format. **Fix:** resize/reformat; **verify current limits in your tenant.**

• **Cause C — Advanced feature without licensing.** Custom branding templates need **Advanced Message

Encryption (E5 / E5 Compliance)**.** Fix:** confirm licensing (R1) or fall back to default branding.

`Set-OMEConfiguration` / `New-TransportRule` — "term is not recognized"

• **Cause:** wrong endpoint. `*-OMEConfiguration`, `*-IRMConfiguration`, `*-TransportRule`,

`Test-IRMConfiguration`, `Get-RMSTemplate` are **Exchange Online PowerShell** (`Connect-ExchangeOnline`). **Fix:** connect Exchange Online; both endpoints ship in `ExchangeOnlineManagement`.

A rule is encrypting more mail than intended after enforcement

• **Cause:** the condition is too broad (e.g. all external recipients) or the scope was widened too

fast. **Fix (immediate):** `Set-TransportRule -Mode Audit` (or `Disable-TransportRule`) — encryption stops at once; tune the condition and re-pilot. Widen scope **group-by-group** with `OME-ScopeHealth.ps1`, re-running the gates each time.

`OME-Report.ps1` counts don't match message trace

• **Cause:** message trace is authoritative; the report is a scoped rollup and message trace does not

label every path as "encrypted." **Fix:** reconcile against message trace and, for Advanced external mail, `Get-OMEMessageStatus` (Gate 9). Message-trace retention is tenant-dependent — **verify.**

Revocation / expiration options aren't available

• **Cause:** message revocation and expiration are **Advanced Message Encryption** features and apply to

OME-portal (external) mail. **Fix:** confirm E5 / E5 Compliance licensing (R1); confirm `Get-OMEMessageStatus` / `Set-OMEMessageRevocation` and the expiration property in your tenant (**verify property/licensing**).

When to escalate beyond this solution

• **What must be encrypted, what triggers Do Not Forward, and how long external mail stays readable** →

compliance / privacy / legal (this solution configures and evidences the method; the mapping is their determination — deferred).

• **Records-retention or legal-hold interaction with encrypted mail** → **records / legal.**

• **Feature genuinely absent despite correct licensing** → verify against current Microsoft

documentation and open a Microsoft support case; OME/Advanced Message Encryption capabilities evolve.

Best Practices — Purview Message Encryption (OME) in Microsoft Purview

Field-tested guidance for running OME well. It follows the same order the solution is built: get the **RMS foundation and branding** right, then let **rules** apply encryption — piloted in `Audit` mode, proportionate to sensitivity, and defensible.

1. Enable and prove the RMS foundation before anything else

- OME is only as healthy as **Azure RMS** underneath it. Confirm `AzureRMSLicensingEnabled : True` and

a clean `Test-IRMConfiguration` **before** you build branding or rules — most "encryption isn't working" tickets trace back to an RMS/IRM problem, not the rule.

- Treat RMS enablement as a **tenant-wide, change-controlled** action; it affects all message

encryption, not just your pilot.

2. Brand the experience — an unbranded portal reads as phishing

- Recipients (especially external ones) are asked to click a link and, often, complete a one-time

passcode. If the wrapper and portal carry **default Microsoft branding or no branding**, people distrust it. Set your **logo, colour, and text** on the default OME configuration so the experience is unmistakably yours.

- Keep the disclaimer and "read the message" text **clear and reassuring** — tell recipients why they

received an encrypted message and what to do. Follow Microsoft's current **logo size/format** limits (verify them; oversized logos fail to render).

3. Always pilot rules in AUDIT mode, scoped to a group

- A mail flow rule with `-Mode Audit` **matches and logs but does not encrypt** — this is your

simulation. Build every OME rule in `Audit` first, scoped to `SG-OME-Pilot`, and confirm the match in **message trace** before flipping to `Enforce`.

- Expand **group-by-group** (or by loosening `-SentToScope` / conditions deliberately), never

tenant-wide in one move. Re-run the Testing gates after each expansion.

4. Pick the template that fits the flow — Encrypt vs Do Not Forward

- `Encrypt` protects the message but lets the recipient **reply and forward** — right when the

recipient needs to work with the content or route it onward within their own controls.

- `Do Not Forward` additionally blocks **forward/print/copy** — reserve it for the most sensitive

one-to-one / one-to-few sends where onward sharing must be prevented. DNF *everything* creates friction and drives workarounds; use it deliberately.

5. Prefer classification-driven triggers over fragile keywords

- A subject keyword like `[encrypt]` is a fine **starting** trigger, but it depends on users

remembering it. As labeling matures, drive encryption from a **sensitivity label** (a `Confidential` / `Highly Confidential` label can apply protection itself or serve as a rule condition) so the strongest protection follows the **classification**, not a typed string.

- Keep the condition→template mapping **narrow and explicit**; over-broad conditions (e.g. "encrypt all

external mail") break automated flows and recipient usability.

6. Set recipient expectations to cut support load

- Document the **recipient experience** for both internal (inline) and external (OTP / OME portal)

correspondents, and give the help desk a short runbook. Most "the encrypted email is broken" tickets are unfamiliarity, not failure.

- Test the **external** experience to your real correspondent domains during the pilot — the OTP/portal

flow varies by provider (Outlook.com, Gmail, etc.).

7. Use Advanced features only where licensed — and know their limits

- **Custom branding templates**, **message revocation** (`Set-OMEMessageRevocation`), and **expiration**

require **Advanced Message Encryption** (E5 / E5 Compliance). Don't design around them on a Basic OME tenant — plan on the **default** branding and the two templates.

- Revocation/expiration apply to mail read via the **OME portal** (typically external recipients);

they don't retroactively control a message a Microsoft 365 recipient already opened inline. **Verify current behaviour and licensing in your tenant.**

8. Don't over-encrypt the plumbing

- Encrypting mail that flows to **systems** (ticketing, journaling, archiving, downstream automation)

that can't process OME will **break** those flows. Exclude or narrowly scope rules away from service/system mailboxes, and confirm journaling/archiving still capture what compliance needs.

9. Keep the evidence trail — message trace and reporting

- **Message trace** (and, for Advanced, `Get-OMEMessageStatus`) is the authoritative record of what was

encrypted. Report **counts by rule/template/time** (`OME-Report.ps1`) for leadership; don't rebuild a shadow log of message contents.

- Retention/disposition of mail-flow and encryption evidence is a **records/legal determination** — do

not delete it to "clean up."

10. Separate the technical control from the sensitivity decision

- OME **enforces** a policy; it does not **decide** what is sensitive. Keep the condition→template

mapping owned by **compliance/privacy**, reviewed as classification and regulation evolve. The administrator operates the control; the business owns the determination (`global-requirements-purview.md` §5).

11. Verify against your tenant — the surfaces evolve

- OME templates, the `*-OMEConfiguration` properties, transport-rule parameters (e.g. the custom-branding

parameter name), and Advanced-feature availability are **licensing-dependent** and the Exchange/Purview portals **change**. Confirm exact paths, parameters, and capability availability **against your licensing/tenant** rather than assuming; treat anything in preview as such.

Deliverables Summary — Purview Message Encryption (OME) in Microsoft Purview

Everything included in this solution, and what each piece is for.

Documentation

Document	Purpose
Solution Overview (`01-solution-overview.md`)	What OME delivers, the RMS foundation, the `Encrypt` / `Do Not Forward` templates, branding, rule-driven encryption, Basic vs Advanced licensing, and the Assess→Design→Configure→Pilot(Audit)→Validate→Expand model.
Requirements (`02-requirements.md`)	OME-specific prerequisites — Basic vs Advanced licensing, Azure RMS enabled/healthy, Exchange admin roles, the branding decision, the trigger→template mapping, optional label triggers, recipient experience, and the pilot/Audit-mode change control.
Implementation (`03-implementation.md`)	Ordered steps: enable RMS, set branding, build the mail flow rules in `Audit` **mode** scoped to a pilot group, validate in a controlled window, then flip to `Enforce` and expand — each object named, with pass/fail gates and rollback.
Testing & Validation (`05-testing.md`)	Ten pass/fail gates from licensing and RMS health through template/branding, Audit-mode detection, a controlled real send, Do-Not-Forward restriction, reporting reconciliation, and the compliance/privacy sign-off before enforcement.

Document	Purpose
Troubleshooting (`06-troubleshooting.md`)	The common OME failure modes (RMS/IRM off, `Test-IRMConfiguration` failing, branding not rendering, rule not matching or matching too much, external recipient can't open, DNF not restricting, Advanced features unavailable) and how to resolve them.
Best Practices (`07-best-practices.md`)	RMS-first, brand the experience, pilot in Audit, right template per flow, classification-driven triggers, recipient expectations, Advanced-only limits, don't over-encrypt system flows, evidence integrity, and separating the control from the sensitivity decision.
Executive Summary (`executive-summary.md`)	One-page business framing for decision-makers.

PowerShell tools (5)

All tools are **read-only or `-WhatIf`/Audit-mode by default**, authenticate **interactively** (no credentials in code), and use **Exchange Online PowerShell** (`Connect-ExchangeOnline`) and/or **Security & Compliance PowerShell** (`Connect-IPPSSession`) / **Microsoft Graph** (`Connect-MgGraph`). Azure RMS activation and the branding **preview** are portal actions; these tools handle discovery, configuration (branding + rules), validation, reporting, and scope/health.

Script	Type	What it does
`OME-Discovery.ps1`	Read-only	OME **readiness/discovery**: reports the IRM/Azure RMS configuration (`AzureRMSLicensingEnabled`), the available RMS templates (`Encrypt` / `Do Not Forward`), existing OME configurations (default + custom branding), any existing mail flow rules that apply/remove OME, and licensing hints (Basic vs Advanced). Changes nothing.

Script	Type	What it does
`OME-Configure.ps1`	`-WhatIf` default	**Configuration/remediation** helper. `-Action Branding` previews/sets the OME branding (`Set-OMEConfiguration`, or `New-OMEConfiguration` for a custom template). `-Action MailFlowRules` prints and (with `-Commit`) creates the `OME-Encrypt-Outbound-Pilot` and `OME-DoNotForward-Confidential` rules **in** `Audit` **mode**, scoped to `SG-OME-Pilot`; a separate `-Enforce` switch is required to flip a rule to `Enforce`. `-WhatIf` by default; nothing changes without `-Commit`.
`OME-Validate.ps1`	Read-only	**Validation checker**: reports whether the OME prerequisites/config are met — RMS enabled, `Test-IRMConfiguration` result, the two rules present with their `Mode` and scope, branding present — as pass/fail lines aligned to `05-testing.md`.
`OME-Report.ps1`	Read-only	**Encryption reporting rollup**: summarizes encrypted-message activity from **message trace** (by rule/template/time) to CSV for a leadership rollup, and — where Advanced Message Encryption applies — reports `Get-OMEMessageStatus` for a supplied message ID. Notes message trace as authoritative; no content/recipient over-collection.
`OME-ScopeHealth.ps1`	Read-only	**Scope/health helper**: runs `Test-IRMConfiguration` (sender/recipient round-trip) and reports RMS/IRM health, the `SG-OME-Pilot` membership and each OME rule's scope, so expansion stays deliberate and reversible; previews the current scope before you widen it.

Named objects (verbatim — used across docs, scripts, and tests)

- **Pilot sender group:** `SG-OME-Pilot`
- **Protection templates:** `Encrypt`, `Do Not Forward` (and user-facing **Encrypt-only**)
- **Default branding object:** the default OME configuration (identity **"OME Configuration"**);

custom template (Advanced only): `OME-Branding-Corp`

- **Mail flow (transport) rules:**
- `OME-Encrypt-Outbound-Pilot` — sender ∈ `SG-OME-Pilot` AND (subject `[encrypt]` OR external

recipient) → template `Encrypt` (Audit first)

- `OME-DoNotForward-Confidential` — message labeled `Highly Confidential` (or agreed

classification) → template `Do Not Forward` (Audit first)

What this solution deliberately does NOT do

- It does **not** decide **what is sensitive** or which mail must be encrypted — that condition→template

mapping is a **compliance/privacy determination** (`global-requirements-purview.md` §5).

- It does **not** enforce encryption before piloting rules in `Audit` **mode** and passing the gates.
- It does **not** invent cmdlets, template names, `*-OMEConfiguration` properties, or transport-rule

parameters — where a name/parameter or capability is licensing- or tenant-specific, it says "verify in your tenant."

- It does **not** assume **Advanced Message Encryption** (custom branding, revocation, expiration) —

those are flagged as E5/E5-Compliance features to verify.

- It does **not** store credentials or secrets; every tool authenticates interactively and defaults to

read-only or `-WhatIf`.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

OME-Configure.ps1

Purpose: Configuration/remediation helper for Purview Message Encryption (OME). With -Action Branding it previews and (with -Commit) sets the OME branding via Set-OMEConfiguration on the default configuration (or New/Set on a custom template for Advanced Message Encryption). With -Action MailFlowRules it prints and (with -Commit) creates the two OME transport rules (OME-Encrypt-Outbound-Pilot -> Encrypt; OME-DoNotForward-Confidential -> Do Not Forward) IN 'Audit' MODE, scoped to the pilot group — the rule matches and logs but DOES NOT ENCRYPT. SAFE BY DEFAULT: prints intended actions under -WhatIf; changes only with -Commit; a rule is flipped to 'Enforce' only when you additionally pass -Enforce. No cmdlets invented; no secrets. HOW TO RUN PowerShell 5.1+ or 7+ with ExchangeOnlineManagement; Connect-ExchangeOnline as an Exchange admin. Preview branding: .\OME-Configure.ps1 -Action Branding -WhatIf Apply branding: .\OME-Configure.ps1 -Action Branding -Commit -EmailText "Your org sent you a protected message." Preview rules: .\OME-Configure.ps1 -Action MailFlowRules -WhatIf Create rules (Audit):.\OME-Configure.ps1 -Action MailFlowRules -Commit -PilotGroup "SG-OME-Pilot@contoso.com" Flip a rule to Enforce (deliberate): .\OME-Configure.ps1 -Action MailFlowRules -Commit -Enforce -RuleName "OME-Encrypt-Outbound-Pilot" EXPECTED RESULTS Branding: Get-OMEConfiguration reflects the values you set. MailFlowRules: the two rules exist, enabled, Mode=Audit (until -Enforce). Nothing changes without -Commit; enforcement requires -Enforce. .NOTES White-label. SAFE-by-default (-WhatIf). Rules created in Audit mode per the category pilot model. Exchange Online PowerShell. No credentials in code. Custom branding templates / Advanced features are licensing-dependent (E5 / E5 Compliance) — verify in your tenant. Determinations about WHAT to encrypt remain with compliance/privacy. No cmdlets invented. >

How to run: PowerShell 5.1+ or 7+ with ExchangeOnlineManagement; Connect-ExchangeOnline as an Exchange admin. Preview branding: .\OME-Configure.ps1 -Action Branding -WhatIf Apply branding: .\OME-Configure.ps1 -Action Branding -Commit -EmailText "Your org sent you a protected message." Preview rules: .\OME-Configure.ps1 -Action MailFlowRules -WhatIf Create rules (Audit):.\OME-Configure.ps1 -Action MailFlowRules -Commit -PilotGroup "SG-OME-Pilot@contoso.com" Flip a rule to Enforce (deliberate): .\OME-Configure.ps1 -Action MailFlowRules -Commit -Enforce -RuleName "OME-Encrypt-Outbound-Pilot" EXPECTED RESULTS Branding: Get-OMEConfiguration reflects the values you set. MailFlowRules: the two rules exist, enabled, Mode=Audit (until -Enforce). Nothing changes without -Commit; enforcement requires -Enforce. .NOTES White-label. SAFE-by-default (-WhatIf). Rules created in Audit mode per the category pilot model. Exchange Online PowerShell. No credentials in code. Custom branding templates / Advanced features are licensing-dependent (E5 / E5 Compliance) — verify in your tenant. Determinations about WHAT to encrypt remain with compliance/privacy. No cmdlets invented. >

OME-Discovery.ps1

Purpose: Connects to Exchange Online PowerShell to assess readiness for Purview Message Encryption (OME). It reports the IRM/Azure RMS configuration (AzureRMSLicensingEnabled), the available RMS protection templates (Encrypt / Do Not Forward), existing OME branding configurations (default + any custom), and any mail flow (transport) rules that apply or remove OME — with a Basic-vs-Advanced licensing hint. Read-only: it changes nothing. HOW TO RUN PowerShell 5.1+ or 7+ with the

ExchangeOnlineManagement module. Connect-ExchangeOnline (MFA-capable Exchange admin), then: .\OME-Discovery.ps1 .\OME-Discovery.ps1 -SaveReport .\OME-Discovery.csv Some settings (Azure RMS activation, branding preview) are portal-managed; where a value has no read surface this script says so. Verify capability/licensing against your tenant (Basic OME via AIP/Azure RMS; Advanced Message Encryption for custom branding / revocation / expiration). EXPECTED RESULTS A readiness summary: RMS enablement (True/False), the RMS templates found, OME configurations, OME-related transport rules with their Mode/State, and a verdict. PASS overall = RMS is enabled and the two protection templates are present. .NOTES White-label. Read-only. Exchange Online PowerShell. No credentials in code. No cmdlets invented. >

How to run: PowerShell 5.1+ or 7+ with the ExchangeOnlineManagement module. Connect-ExchangeOnline (MFA-capable Exchange admin), then: .\OME-Discovery.ps1 .\OME-Discovery.ps1 -SaveReport .\OME-Discovery.csv Some settings (Azure RMS activation, branding preview) are portal-managed; where a value has no read surface this script says so. Verify capability/licensing against your tenant (Basic OME via AIP/Azure RMS; Advanced Message Encryption for custom branding / revocation / expiration). EXPECTED RESULTS A readiness summary: RMS enablement (True/False), the RMS templates found, OME configurations, OME-related transport rules with their Mode/State, and a verdict. PASS overall = RMS is enabled and the two protection templates are present. .NOTES White-label. Read-only. Exchange Online PowerShell. No credentials in code. No cmdlets invented. >

OME-Report.ps1

Purpose: Read-only reporting for Purview Message Encryption (OME). Uses message trace (Get-MessageTraceV2 / Get-MessageTrace) over a bounded window to produce a defensible rollup of mail that was processed by the OME rules — counts by sender/recipient-domain/day — without exposing message bodies. Maps to Testing Gate 9 (reporting reconciles with message trace). Message trace is authoritative; this is a summarizing convenience. HOW TO RUN PowerShell 5.1+ or 7+ with ExchangeOnlineManagement; Connect-ExchangeOnline as an Exchange admin. .\OME-Report.ps1 -Days 7 .\OME-Report.ps1 -Days 2 -SenderDomain contoso.com -SaveReport .\OME-Report.csv EXPECTED RESULTS A per-day / per-recipient-domain rollup of outbound messages in the window for the pilot senders, which you reconcile against message trace and (for Advanced external mail) Get-OMEMessageStatus. .NOTES White-label. Read-only (no -Commit path). Exchange Online PowerShell. No credentials in code. Message trace does not label "encrypted" directly for every path — reconcile with the rule Audit results / Get-OMEMessageStatus. Retention of message trace is tenant-dependent — verify. No cmdlets invented. >

How to run: PowerShell 5.1+ or 7+ with ExchangeOnlineManagement; Connect-ExchangeOnline as an Exchange admin. .\OME-Report.ps1 -Days 7 .\OME-Report.ps1 -Days 2 -SenderDomain contoso.com -SaveReport .\OME-Report.csv EXPECTED RESULTS A per-day / per-recipient-domain rollup of outbound messages in the window for the pilot senders, which you reconcile against message trace and (for Advanced external mail) Get-OMEMessageStatus. .NOTES White-label. Read-only (no -Commit path). Exchange Online PowerShell. No credentials in code. Message trace does not label "encrypted" directly for every path — reconcile with the rule Audit results / Get-OMEMessageStatus. Retention of message trace is tenant-dependent — verify. No cmdlets invented. >

OME-ScopeHealth.ps1

Purpose: Read-only health + scope helper for Purview Message Encryption (OME). Runs Test-IRMConfiguration for a sender/recipient pair to confirm the RMS encrypt/decrypt round-trip (Testing Gate 2), and previews the pilot group's current membership so an expansion is staged deliberately (group-by-group) rather than tenant-wide. It only reads and tests — it never adds members or changes rules. HOW TO RUN PowerShell 5.1+ or 7+ with ExchangeOnlineManagement; Connect-ExchangeOnline as an Exchange admin. .\OME-ScopeHealth.ps1 -Sender pilot@contoso.com -Recipient external@example.com .\OME-ScopeHealth.ps1 -PilotGroup "SG-OME-Pilot@contoso.com" -Sender pilot@contoso.com -Recipient external@example.com EXPECTED RESULTS Test-IRMConfiguration returns PASS on all checks; the pilot group's members are listed so you can see who a rule change would affect before you widen scope. .NOTES White-label. Read-only (no membership/rule changes). Exchange Online PowerShell. No credentials in code. Expansion (adding members / broadening -FromMemberOf) is a deliberate, gated step (03-implementation.md Phase 5). No cmdlets invented. >

How to run: PowerShell 5.1+ or 7+ with ExchangeOnlineManagement; Connect-ExchangeOnline as an Exchange admin. .\OME-ScopeHealth.ps1 -Sender pilot@contoso.com -Recipient external@example.com .\OME-ScopeHealth.ps1 -PilotGroup "SG-OME-Pilot@contoso.com" -Sender pilot@contoso.com -Recipient external@example.com EXPECTED RESULTS Test-IRMConfiguration returns PASS on all checks; the pilot group's members are listed so you can see who a rule change would affect before you widen scope. .NOTES White-label. Read-only (no membership/rule changes). Exchange Online PowerShell. No credentials in code. Expansion (adding members / broadening -FromMemberOf) is a deliberate, gated step (03-implementation.md Phase 5). No cmdlets invented. >

OME-Validate.ps1

Purpose: Read-only validation for Purview Message Encryption (OME). Confirms the RMS foundation, that the expected OME branding configuration exists, and that the two mail flow rules (OME-Encrypt-Outbound-Pilot / OME-DoNotForward-Confidential) exist, are ENABLED, are in Mode=Audit, and are scoped to the pilot group (SG-OME-Pilot). Maps directly to Testing Gate 5. Changes nothing. HOW TO RUN PowerShell 5.1+ or 7+ with ExchangeOnlineManagement; Connect-ExchangeOnline as an Exchange admin. .\OME-Validate.ps1 .\OME-Validate.ps1 -PilotGroup "SG-OME-Pilot@contoso.com" -SaveReport .\OME-Validate.csv EXPECTED RESULTS PASS = both rules present, enabled, Mode=Audit, scoped to the pilot group, each with a rights- protection action. A rule that is already Enforce, disabled, or over-scoped is flagged so it is corrected before enforcement. .NOTES White-label. Read-only. Exchange Online PowerShell. No credentials in code. No cmdlets invented. Enforcing/expanding a rule remains a deliberate, gated step (03-implementation.md Phase 5). >

How to run: PowerShell 5.1+ or 7+ with ExchangeOnlineManagement; Connect-ExchangeOnline as an Exchange admin. .\OME-Validate.ps1 .\OME-Validate.ps1 -PilotGroup "SG-OME-Pilot@contoso.com" -SaveReport .\OME-Validate.csv EXPECTED RESULTS PASS = both rules present, enabled, Mode=Audit, scoped to the pilot group, each with a rights- protection action. A rule that is already Enforce, disabled, or over-scoped is flagged so it is corrected before enforcement. .NOTES White-label. Read-only. Exchange Online PowerShell. No credentials in code. No cmdlets invented. Enforcing/expanding a rule remains a deliberate, gated step (03-implementation.md Phase 5). >